

Python Assignment-22

1. Write a program that accepts a list of integers and uses `pool.map()` to calculate the sum of squares from 1 to N for every element in the list.

Example Input

[1000000,2000000,3000000,4000000]

Expected Output

[333333833333500000,
2666668666667000000,

...

]

Answer :

```
from multiprocessing import Pool

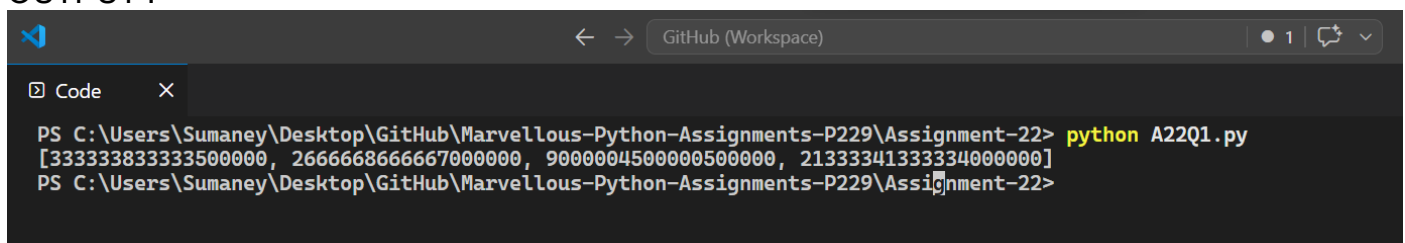
def square(x):
    Total = 0
    for i in range(1,x+1):
        Total = Total + (i * i)
    return Total

def main():
    numbers = [1000000,2000000,3000000,4000000]

    with Pool(4) as p:
        result = p.map(square, numbers)
        print(result)

if __name__ == "__main__":
    main()
```

OUTPUT :



```
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22> python A22Q1.py
[333333833333500000, 2666668666667000000, 9000004500000500000, 2133334133334000000]
```

2. Write a program that calculates factorials of multiple numbers simultaneously using `Pool.map()`.

Input [10,15,20,25]

Display

- Process ID
- Input Number
- Factorial

Answer :

```
from multiprocessing import current_process
from multiprocessing import Pool

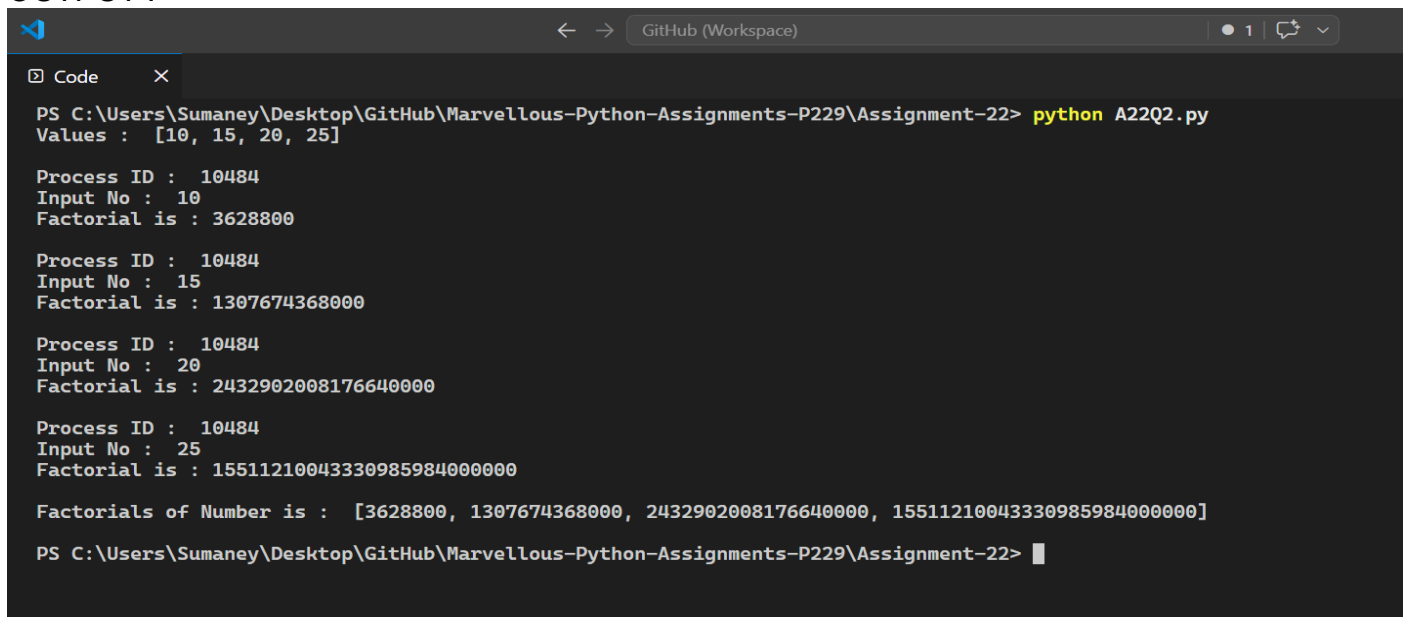
def Factorial(Values):
    fact = 1
    for value in range(1,Values+1):
        fact = fact * value
    print(Count(Values))
    print("Process ID : ",current_process().pid)
    print("Input No : ",Values)
    print("Factorial is :",fact)
    print()
    return fact

def main():
    Values = [10,15,20,25]
    print("Values : ",Values)
    print()

    with Pool(4) as p:
        Result = p.map(Factorial,Values)
        print("Factorials of Number is : ",Result)
        print()

if __name__ == "__main__":
    main()
```

OUTPUT :



```
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22> python A22Q2.py
Values : [10, 15, 20, 25]

Process ID : 10484
Input No : 10
Factorial is : 3628800

Process ID : 10484
Input No : 15
Factorial is : 1307674368000

Process ID : 10484
Input No : 20
Factorial is : 2432902008176640000

Process ID : 10484
Input No : 25
Factorial is : 15511210043330985984000000

Factorials of Number is : [3628800, 1307674368000, 2432902008176640000, 15511210043330985984000000]
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22>
```

3. For every number in the given list, count how many prime numbers exist between 1 and N using multiprocessing Pool.

INPUT : 10000 20000 30000 40000 OUTPUT : Display total prime count for each number.

Answer :

```
from multiprocessing import Pool

def Prime(Value):
    PrimeCount = 0
    for Number in range(1, Value+1):
        Count = 0
        for i in range(1, Number+1):
            if Number%i==0:
                Count = Count + 1
        if Count == 2:
            PrimeCount = PrimeCount + 1
    return PrimeCount

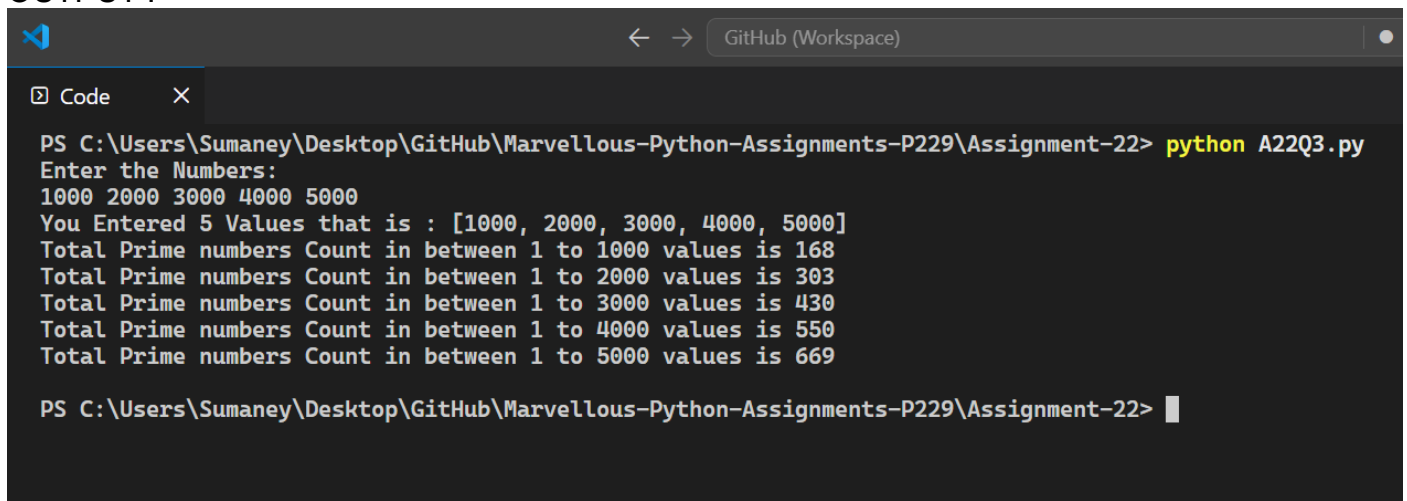
def main():
    print("Enter the Numbers: ")
    Values = list(map(int, input().split()))
    print(f"You Entered {len(Values)} Values that is : {Values}")

    with Pool(4) as p:
        Result = p.map(Prime, Values)

    for Value, Count in zip(Values, Result):
        print(f"Total Prime numbers Count in between 1 to {Value} values is {Count}")
    print()

if __name__ == "__main__":
    main()
```

OUTPUT :



```
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22> python A22Q3.py
Enter the Numbers:
1000 2000 3000 4000 5000
You Entered 5 Values that is : [1000, 2000, 3000, 4000, 5000]
Total Prime numbers Count in between 1 to 1000 values is 168
Total Prime numbers Count in between 1 to 2000 values is 303
Total Prime numbers Count in between 1 to 3000 values is 430
Total Prime numbers Count in between 1 to 4000 values is 550
Total Prime numbers Count in between 1 to 5000 values is 669

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22>
```

4. Write a program that calculates $1^5+2^5+3^5+.....+N^5$ for multiple values of N simultaneously using Pool.

INPUT : [1000000, 2000000, 3000000, 4000000]

OUTPUT : Measure total execution time.

Answer :

```
from multiprocessing import Pool
import time

def Power(iValues):
    Sum = 0
    for Value in range(1,iValues+1):
        Sum = Sum + (Value ** 5)
    return Sum

def main():
    StartTime = time.time()

    print("Enter the list of Numbers : ")
    Values = list(map(int,input().split()))
    print("You Entered list is :",Values)

    with Pool(4) as p:
        Result = p.map(Power,Values)

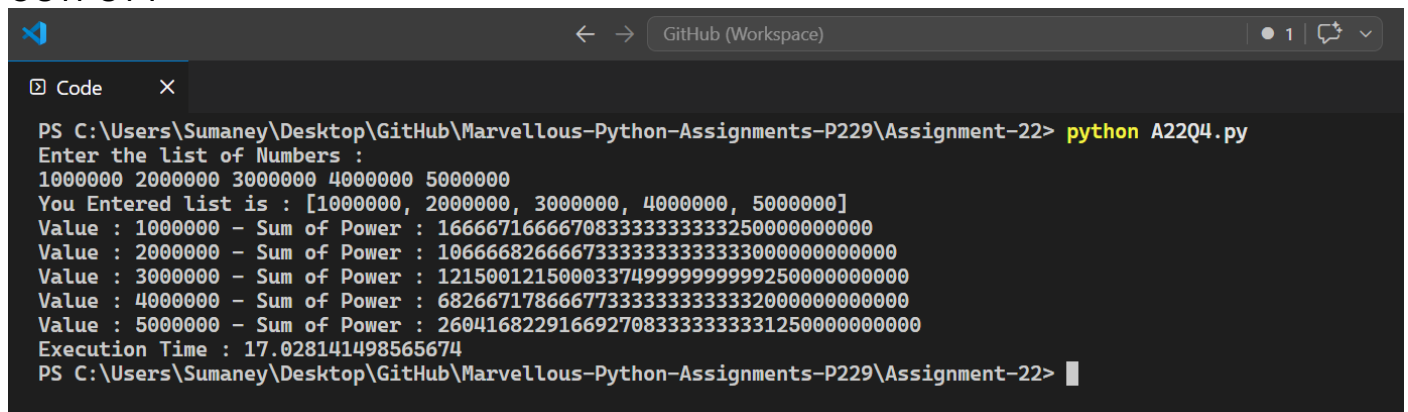
    for Value,PowerSum in zip(Values,Result):
        print(f"Value : {Value} - Sum of Power : {PowerSum} ")

    EndTime = time.time()

    print(f"Execution Time : {EndTime - StartTime}")

if __name__ == "__main__":
    main()
```

OUTPUT :



```
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22> python A22Q4.py
Enter the list of Numbers :
1000000 2000000 3000000 4000000 5000000
You Entered list is : [1000000, 2000000, 3000000, 4000000, 5000000]
Value : 1000000 - Sum of Power : 1666671666670833333333333333250000000000
Value : 2000000 - Sum of Power : 1066668266667333333333333333300000000000
Value : 3000000 - Sum of Power : 121500121500033749999999999250000000000
Value : 4000000 - Sum of Power : 6826671786667733333333333333200000000000
Value : 5000000 - Sum of Power : 26041682291669270833333333331250000000000
Execution Time : 17.028141498565674
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-22>
```

